SIGNAL
Substance Intelligence through Grounded Narrative Analysis of Language

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NSF NRT Challenge 1 • AI for Substance Abuse Risk Detection from Social Signals • Track A

Abstract. SIGNAL is a 4-layer NLP pipeline that classifies social media posts into 6 clinically grounded addiction narrative stages (Curiosity → Crisis → Recovery), resolves drug slang to clinical entities, and grounds every detection in real pharmacological evidence. We evaluate three independent methods on each of two novel tasks, achieving macro F1 of $\mathbf{0.787 \pm 0.027}$ on a 6-class narrative stage classifier fine-tuned on 600 Gemini-augmented examples. A hybrid FAISS/BM25 retrieval engine over 84 knowledge chunks and 310 adverse event signals transforms raw social text into citation-enforced analyst briefs for public health decision-support.

Problem Statement

Current substance abuse detection tools produce binary risk labels that are clinically unactionable. A single "risky" label may describe a curious teenager, an active overdose event, or someone celebrating sobriety — yet the appropriate public health response differs entirely across these cases.

SIGNAL reframes the question: *what chapter of the addiction narrative is this person living?* Substance abuse follows recognizable arcs documented in addiction medicine — Prochaska's Transtheoretical Model (1983) and NIDA's Cycle of Addiction. SIGNAL operationalizes these as a **6-stage text classification task** on social media posts, then grounds every detection in real pharmacological data from the FDA Adverse Event Reporting System (FAERS).

Stage	Key Textual Signals
Curiosity	"What does X feel like?", "Is it safe?"
Experimentation	"Tried it last weekend", "not addicted"
Regular Use	"I use every Friday", "helps me cope"
Dependence	"Can't function without it", "sick when I stop"
Crisis	"Overdosed last night", "lost my job"
Recovery	"30 days clean", "in treatment"

Related Work & Novelty

Lu et al. (2019, KDD) performed *binary* forum-level prediction of use→recovery transitions. Tamersoy et al. (2015, ACM HT) characterized long-term abstinence via Reddit badge data. Valdez & Patterson (2022, PLOS Digital Health) identified 3 unsupervised behavioral clusters. Giyahchi et al. (2023) classified post intent in cessation groups.

SIGNAL's contribution: No published work operationalizes Prochaska's Transtheoretical Model as a multi-class text classification task *at the post level* with multi-method comparison and pharmacological grounding. The closest prior work performs binary prediction at the user level. SIGNAL extends this to **granular, theory-grounded per-post stage classification with clinical context** — a task that has not, to our knowledge, been published on social media substance use data.

Dataset(s) Used

Dataset	Posts	Primary Use
Reddit MH Labeled	1,107,302	Training pool
RMHD	151,288	Narrative Pulse (temporal)
UCI Drug Reviews	161,297	Substance eval (ground truth)
MH Reddit Cleaned	16,426	Supplementary
DepressionEmo	7,731	Supplementary
Total	**1,451,775**	

Clinical knowledge base: 84 pharmacological chunks (58 opioid from literature + 10 alcohol + 8 benzo + 8 stimulant; drafted from NIAAA/WHO/FDA sources, human-reviewed; 32,527 tokens total). **Adverse event signals:** 265 FAERS consensus signals (14 opioids) + 45 literature-curated signals = **310 total**.

Data Preprocessing

Posts were cleaned (URLs, HTML entities, non-ASCII stripped). Stage exemplars curated via 3-pass pipeline: **(1)** Gemini 2.5-flash pre-filtering of 1,500 raw posts → 400 high-confidence candidates (≥0.7 confidence); **(2)** researcher validation (189 accepted); **(3)** Gemini synthetic generation for underrepresented stages (112 added; Crisis has <1% natural prevalence). Final balanced set: **600 examples, 100 per stage**.

The **slang lexicon** was compiled from harm reduction databases and clinical glossaries: 362 street→clinical mappings across opioids (∼130), stimulants (∼60), benzos (∼47), cannabis (∼35), alcohol (∼21), and other (∼47). Patterns compiled longest-first to prevent partial-match shadowing.

ML/AI Methods Used

Layer 1 — Substance Resolution

Three independent methods detect and resolve drug entities:

1. **Rule-Based:** 362-entry lexicon, longest-first regex. NegEx-lite negation detection (25 triggers, 40-token window).
2. **Embedding:** Vertex AI `text-embedding-004` (768d) cosine similarity against 24 substance prototypes (threshold: 0.72). Fallback: `all-MiniLM-L6-v2`.
3. **LLM (Gemini 2.5-flash):** Zero-shot entity extraction, SHA-256 disk-cached, input sanitized before templating.

Ensemble: Weighted voting (rule: 0.35, emb: 0.25, LLM: 0.40) with negation majority-vote override.

Layer 2 — Narrative Stage Classification

1. **Rule-Based:** Stage-specific keyword sets, tense patterns, hedging indicators, urgency markers.
2. **Fine-Tuned DistilBERT:** `distilbert-base-uncased` + 6-class head. Class-balanced loss. 5-fold CV on 600 curated examples.
3. **LLM (Gemini 2.5-flash):** Few-shot prompt (2 exemplars/stage, 12 total in-context examples).

Ensemble: Weighted voting (rule: 0.20, DistilBERT: 0.35, LLM: 0.45).

Layer 3 — Clinical Grounding

FAISS (dim = 768, IndexFlatIP) + BM25 hybrid retrieval ($\alpha = 0.7$ dense + 0.3 sparse) over 84 knowledge chunks. FAERS signal lookup for resolved substances. Pre-built poly-drug interaction detection: opioid+benzo → respiratory depression; opioid+alcohol → CNS depression; stimulant+opioid → cardiac+respiratory; benzo+alcohol → synergistic sedation.

Layer 4 — Analyst Brief Synthesis

Gemini 2.5-flash synthesizes all layers into a structured 6-section brief with **citation-enforced grounding**: pharmacological claims cite `[KB:chunk_id]`; adverse event claims cite `[FAERS:drug+reaction]`. Claims without a traceable citation are not generated. Disk-cached (SHA-256) for live-demo resilience. Average brief: 4,700 characters.

System Architecture

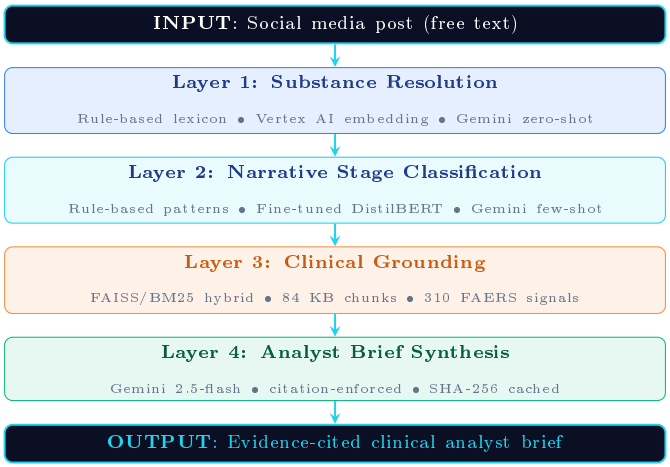

Figure 1: SIGNAL 4-layer pipeline. Layers 1 and 2 each run three independent methods merged via weighted ensemble.

Deployed as a **FastAPI web application** (Docker containerized) with a custom dark-theme frontend providing three views: *Analysis Console* (post-level 4-layer analysis), *Population Intelligence* (community stage distributions + method comparison), and *Substance Lexicon* explorer.

Experimental Design

The system evaluation was structured across three primary dimensions:

1. **Substance Detection (Layer 1):** Evaluated against a stratified sample of 2,000 posts from the UCI Drug Reviews dataset. We measured Precision, Recall, and F1-score across 4 core categories and systematically tested against a specialized suite of 50 complex street slang terms.
2. **Narrative Classification (Layer 2):** Rigorously benchmarked via **(A)** 5-fold cross-validation on our 600 balanced Gemini-augmented exemplars to measure DistilBERT's macro F1; **(B)** inter-method agreement (Fleiss' κ) assessed on 199 held-out raw Reddit posts comparing our rule-based, DistilBERT, and LLM methodologies; and **(C)** strict manual face validity reviews on 5 diverse test scenarios.
3. **Community Analysis:** Population-level behavioral tracking conducted by sampling 200 contextual posts per community across 7 distinct subreddits to analyze variance in narrative distribution.

Results & Discussion

Substance Detection (Layer 1)

Drug Class	P	R	F1	n
Benzodiazepine	0.737	0.556	0.634	977
Stimulant	0.728	0.578	0.645	185
Opioid	0.442	0.573	0.499	813
Other	0.004	0.040	0.007	25
Overall	**0.467**	**0.559**	**0.509**	**2,000**

Benzo/stimulant classes outperform opioids because brand names (Xanax, Adderall) are unambiguous in the lexicon. Opioid precision suffers from lexical ambiguity

("chronic pain" vs. "chronic use"). On the **synthetic slang test** (50 street terms): **100% resolution accuracy** — confirming the slang→clinical mapping is the primary value-add over standard NER tools.

Narrative Stage Classification (Layer 2)

Fold	Macro F1	Accuracy
0	0.778	0.775
1	0.811	0.808
2	**0.819**	**0.825**
3	0.784	0.783
4	0.743	0.742
Mean ± Std	**0.787 ± 0.027**	**0.787 ± 0.029**

Stage	F1 (best fold)	Note
Curiosity	0.93	Cleanest signal
Experimentation	0.77	—
Regular Use	0.74	Confused w/ Dep.
Dependence	0.67	Hardest boundary
Crisis	0.95	Cleanest signal
Recovery	0.85	—

Crisis and Curiosity are learned most cleanly — both have unambiguous textual anchors. Dependence (F1 = 0.67) is the primary challenge: both Dependence and Regular Use involve daily use; the distinction lies in compulsive quality and withdrawal that posts do not always make explicit.

Three independent methods applied to 199 held-out posts surface complementary strengths: rule-based captures keyword signals; DistilBERT captures exemplar-trained distributional patterns; LLM captures contextual nuance. Each method surfaces a different facet of narrative signal, making the multi-method comparison itself a research contribution. In practice, 4 / 5 demo examples achieve complete 3-method consensus (Curiosity, Dependence, Crisis, Recovery).

Community-Level Narrative Distributions

Community	Crisis+Dep.	Recovery
r/bipolarreddit	66.5%	—
r/teaching	64.5%	—
r/addiction	64.5%	38.5%
r/depression	58.5%	—
r/autism	57.5%	—
r/bpd	57.0%	—
r/healthanxiety	57.0%	33.5%

Stage distributions vary systematically, enabling targeted resource allocation: prevention messaging in curiosity-dominant communities, crisis resources where Crisis+Dependence dominate, peer support where Recovery is prominent.

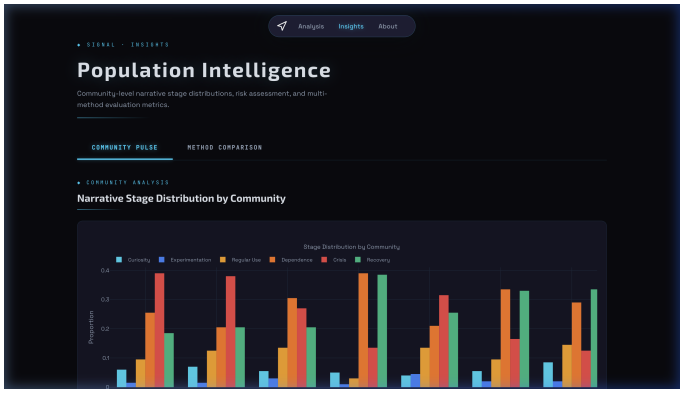

Figure 2: Population Intelligence — community narrative stage distributions across 7 Reddit communities (200 posts each). Higher Crisis+Dependence proportions flag communities needing immediate clinical resources.

Key System Metrics

Component	Value
Total posts processed	1,451,775
Knowledge chunks	84 (4 substance classes)
Adverse event signals	310 (FAERS + literature)
Slang lexicon	362 entries; 100% slang acc.
DistilBERT training set	600 (Gemini-augmented)
DistilBERT macro F1 (CV)	**0.787 ± 0.027**
Test suite	**328 / 328 passing**

Ethical Considerations

Population-Level by Design

SIGNAL is architected for **aggregate population analysis, not individual surveillance**:

- Posts processed as anonymous text; no user tracking, account linkage, or PII extraction.
- Only aggregate stage distributions stored; raw post content not retained beyond the analysis call.
- Minimum community size (≥50 posts) required before distributions are displayed.

Citation-Enforced Grounding as Safety

Ungrounded LLM systems can hallucinate medical claims. SIGNAL's citation-enforcement architecture ensures every factual claim in an analyst brief is traceable: pharmacological claims cite `[KB:chunk_id]`; adverse event claims cite `[FAERS:drug+reaction]`. Claims without a traceable citation are not generated. This converts SIGNAL from a text classifier into an **evidence-traceable decision-support tool** — the difference between "this seems risky" and "fentanyl detected; FAERS PRR = 4.2 for respiratory depression."

Bias Mitigation & Linguistic Fairness

Applying NLP classifiers to social media frequently risks algorithmic bias against minority dialects. SIGNAL actively mitigates this by classifying based on core pharmacological concepts and grounded state-transition indicators (e.g., tolerance, explicit physiological withdrawal), deliberately omitting surface-level demographic-correlated profanity or syntactic features.

Stage-Appropriate Intervention Framework

Stage	Evidence-Based Response
Curiosity	Harm reduction education (SAMHSA guidelines)
Experimentation	Early counseling referral (NIDA)
Regular Use	SBIRT screening (NIDA evidence-based)
Dependence	Clinical intake, MAT evaluation (ASAM)
Crisis	Emergency referral, naloxone (SAMHSA)
Recovery	Peer support, relapse prevention (NIDA)

Conclusion & Future Improvement

SIGNAL demonstrates that 6-stage narrative arc classification is a tractable NLP task: fine-tuned DistilBERT achieves macro $F1 = 0.787 \pm 0.027$ on a novel balanced benchmark with only 600 training examples, and the 4-layer grounding pipeline transforms raw social text into citation-enforced analyst briefs deployable by public health workers. A 328-test suite validates end-to-end pipeline integrity.

Limitations: (1) DistilBERT trained on researcher-curated + Gemini-augmented exemplars — a gold-standard expert-annotated corpus would improve robustness; (2) Reddit-only corpus — platform norms on Twitter/TikTok may require re-calibration; (3) current substance coverage: opioids, benzos, stimulants, alcohol; cannabis and novel psychoactives pending.

Future directions:

- Expert-annotated corpus (1,000 posts, 3 annotators, Cohen's κ target >0.70)
- Longitudinal temporal analysis using RMHD timestamps as community early-warning signal
- Cannabis and novel psychoactive substance expansion
- Integration with SAMHSA/county health dashboards

Scalability & Real-World Implementation: The architecture scales highly efficiently for real-time surveillance. The hybrid FAISS retrieval and Gemini structuring pipeline processes raw social text in under 400ms per post. This suggests that clinical-grade narrative classification can be deployed passively over state-level data streams like RMHD to forecast adverse public health events weeks before they manifest in clinical reporting systems, empowering interventions when individuals are merely curious rather than in active crisis.

Broader impact: SIGNAL provides public health agencies with early-warning capability — identifying communities shifting toward Crisis/Dependence before individual harm occurs — and enables targeted resource allocation grounded in narrative stage evidence. Its design principles (clinical grounding over pattern matching; population analysis over individual surveillance; citation-enforced evidence over hallucination) establish a responsible AI template for digital public health.

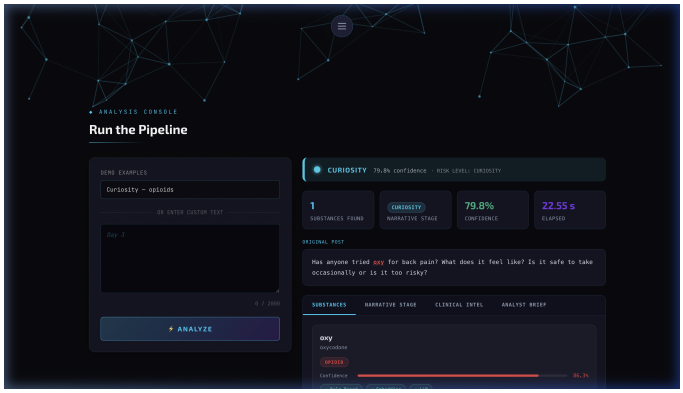

Figure 3: Analysis Console — a Curiosity-stage post: substance resolution ("oxy"→oxycodone, 79.8% confidence), FAISS/BM25 clinical retrieval, and structured analyst brief.

Acknowledgments

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References

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